



Instituto Politécnico Nacional

Escuela Superior de Cómputo

Selected Topics in Cryptography

Reporte Práctica 01: “Non-singular elliptic curves”

Alumno:

Sigala Morales Said

2022630413

Grupo: 7CM4

Maestro: Sandra Diaz Santiago

Fecha de realización: 19 de febrero de 2025

Fecha de entrega: 25 de febrero de 2025



PSEUDOCÓDIGO

1. Write the pseudocode of an algorithm that receives as input, the parameters of a non-singular elliptic curve a, b, p , where $p \geq 3$ is a prime number. The output must be the points that belong to curve, using a notation of three coordinates, i.e the point of infinity $\mathcal{O} = (0, 1, 0)$ and for any other point $P \neq \mathcal{O}$, $P = (x, y, 1)$. Please use the notation for pseudocode given in class.

Algoritmo 1: Encontrar los puntos en un Curva elíptica.

Input: Tres números enteros que describen una curva no singular a, b, p donde p es un primo mayor a 2.

Output: Conjunto de los puntos que conforman una curva en notación de tres coordenadas.

```
1. puntos  $\leftarrow \{(0, 1, 0)\}$ 
2. for  $x \leftarrow 0$  to  $p - 1$  do
3.    $pec \leftarrow (x^3 + a * x + b) \bmod p$ 
4.   for  $y \leftarrow 0$  to  $p - 1$  do
5.     if  $(y^2 \bmod p) == pec$  then
6.        $p \leftarrow (x, y, 1)$ 
7.       agregar  $p$  a puntos
8.     end
9.   end
10. end
11. return puntos
```



PROGRAMMING EXERCISES

1. Design a function that receives as input a prime number $p \geq 3$ to find the quadratic residues modulo p and also the square roots modulo p . For example if $p = 11$ and we know that $2^2 \bmod 11 = 4$ and $9^2 \bmod 11 = 4$ we know that 4 is a quadratic residue and its square roots are 2 and 9.

```
1 def residuos_cuadraticos(p):
2
3     if p < 2 or not isprime(p):
4         raise ValueError("El número debe ser primo y mayor a 2")
5     residuos = {}
6     for x in range(p):
7         residuo = (x * x) % p
8         if residuo not in residuos:
9             residuos[residuo] = []
10            residuos[residuo].append(x)
11
12     return residuos
13
```

Explicación de la función: En el if se verifica que sea mayor que 2 el primo que se le da sino manda una interrupción. Posteriormente se declara un array donde se almacenarán los residuos cuadráticos, el for posterior hace el calculo de los residuos, realiza la formula y guarda los resultados en el array para al ultimo retornar ese array una vez que todo el for haya terminado.



2. Design a function that receives as input a prime number $p \geq 3$, and generates a non-singular elliptic curve over $\mathbb{Z}_p$, ie. your function must output the parameters a, b such that the discriminant is different from zero.

```
1 def generar_ec_no_singular(p):
2     if p < 2 or not isprime(p):
3         raise ValueError("El número debe ser primo y mayor a 2")
4
5     while True:
6         a = random.randint(1, p - 1)
7         b = random.randint(1, p - 1)
8         if (4 * pow(a, 3) + 27 * pow(b, 2)) % p != 0:
9             return a, b, f'y^2 = x^3 + {a}x + {b}'
```

Descripción de la función:

Verifica que p sea primo y mayor a 2, posteriormente en el while al azar escoge 2 valores de a y b entre 1 y $1-p$ tal que al calcular el discriminante este sea distinto de 0. Si se comprueba esto procedemos a armar la ecuación de la curva y la retornamos.



3. Design a function that receives as input the parameters of a non-singular elliptic curve E , given by a, b , a prime number $p \geq 3$ and a point of three coordinates $P = (x, y, 1)$ where $x, y \in \mathbb{Z}_p$ or $\mathcal{O} = (0, 1, 0)$. The output of your function will be true if $P \in E$ or false if $P \notin E$

```
1 def is_point_on_curve(a, b, p, P):
2     x, y, z = P
3
4     # Caso del punto en el infinito
5     if (x, y, z) == (0, 1, 0):
6         return True
7
8
9     lhs = (y ** 2) % p
10    rhs = (x ** 3 + a * x + b) % p #r
11    return lhs == rhs
```

Se le pasa a la función todo lo necesario para que pueda armar la curva elíptica, así como el punto a evaluar con sus tres coordenadas.

Lo primero que hace la función es descomponer el punto en sus tres coordenadas, evalúa que no sea el punto al infinito, de ser así devolvería true automáticamente. Sino es el punto al infinito sustituye los valores en ambos lados de la ecuación y de ser iguales se devuelve true.



5 CASOS DE PRUEBA

P: 59

```
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01> & C:\Users\saiids\AppData\Local\Microsoft\WindowsApps\python3.12.exe "c:/Users/saiids/Documents/ESCOM/BvoSemestre/CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio3.py"
Residuos cuadráticos y raíces cuadradas para 59: [0: [0], 1: [1, 58], 4: [2, 57], 9: [3, 56], 16: [4, 55], 25: [5, 54], 36: [6, 53], 49: [7, 52], 5: [8, 51], 22: [9, 50], 41: [10, 49], 3: [11, 48], 26: [12, 47], 51: [13, 46], 19: [14, 45], 48: [15, 44], 20: [16, 43], 53: [17, 42], 29: [18, 41], 7: [19, 40], 46: [20, 39], 28: [21, 38], 12: [22, 37], 57: [23, 36], 45: [24, 35], 35: [25, 34], 27: [26, 33], 21: [27, 32], 17: [28, 31], 15: [29, 30]]
Curva elíptica no singular para 59:  $y^2 = x^3 + 23x + 46$ 
Valor de a: 23
Valor de b: 46
Puntos en la curva: [(0, 20), (0, 39), (2, 10), (2, 49), (4, 5), (4, 54), (6, 20), (6, 39), (7, 14), (7, 45), (13, 8), (13, 51), (15, 7), (15, 52), (16, 12), (16, 47), (19, 12), (19, 47), (22, 7), (22, 52), (23, 23), (23, 36), (24, 12), (24, 47), (25, 27), (25, 32), (26, 15), (26, 44), (28, 24), (28, 35), (29, 26), (29, 33), (34, 22), (34, 37), (35, 19), (35, 40), (36, 25), (36, 34), (40, 19), (40, 40), (43, 19), (43, 40), (45, 15), (45, 44), (46, 21), (46, 38), (47, 15), (47, 44), (53, 20), (53, 39), (56, 3), (56, 56), (57, 13), (57, 46), (58, 9), (58, 50)]
El punto (0, 20, 1) está en la curva: True
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01>
```

P: 71

```
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01> & C:\Users\saiids\AppData\Local\Microsoft\WindowsApps\python3.12.exe "c:/Users/saiids/Documents/ESCOM/BvoSemestre/CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio3.py"
Residuos cuadráticos y raíces cuadradas para 71: [0: [0], 1: [1, 70], 4: [2, 69], 9: [3, 68], 16: [4, 67], 25: [5, 66], 36: [6, 65], 49: [7, 64], 64: [8, 63], 10: [9, 62], 29: [10, 61], 50: [11, 60], 2: [12, 59], 27: [13, 58], 54: [14, 57], 12: [15, 56], 43: [16, 55], 5: [17, 54], 48: [18, 53], 6: [19, 52], 45: [20, 51], 15: [21, 50], 58: [22, 49], 32: [23, 48], 8: [24, 47], 57: [25, 46], 37: [26, 45], 19: [27, 44], 3: [28, 43], 60: [29, 42], 48: [30, 41], 38: [31, 40], 30: [32, 39], 24: [33, 38], 20: [34, 37], 18: [35, 36]]
Curva elíptica no singular para 71:  $y^2 = x^3 + 7x + 24$ 
Valor de a: 7
Valor de b: 24
Puntos en la curva: [(0, 33), (0, 38), (1, 23), (1, 48), (3, 1), (3, 70), (4, 20), (4, 51), (8, 33), (8, 38), (10, 10), (10, 61), (11, 15), (11, 56), (13, 18), (13, 53), (15, 5), (15, 66), (16, 16), (16, 55), (17, 21), (17, 50), (18, 35), (18, 36), (19, 22), (19, 49), (21, 29), (21, 42), (24, 10), (24, 61), (26, 23), (26, 48), (27, 4), (27, 67), (28, 34), (28, 37), (29, 11), (29, 60), (32, 1), (32, 70), (34, 27), (34, 44), (36, 1), (36, 70), (37, 10), (37, 61), (40, 7), (40, 64), (44, 23), (44, 48), (45, 4), (45, 67), (46, 25), (46, 46), (47, 27), (47, 44), (48, 11), (48, 60), (51, 7), (51, 64), (53, 32), (53, 39), (55, 17), (55, 54), (58, 24), (58, 47), (59, 22), (59, 49), (60, 6), (60, 65), (61, 27), (61, 44), (63, 33), (63, 38), (64, 22), (64, 49), (65, 11), (65, 60), (66, 19), (66, 52), (67, 28), (67, 43), (69, 12), (69, 59), (70, 4), (70, 67)]
El punto (0, 33, 1) está en la curva: True
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01>
```

P: 53

```
ments/ESCOM/BvoSemestre/CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio3.py"
Residuos cuadráticos y raíces cuadradas para 53: [0: [0], 1: [1, 52], 4: [2, 51], 9: [3, 50], 16: [4, 49], 25: [5, 48], 36: [6, 47], 49: [7, 46], 11: [8, 45], 28: [9, 44], 47: [10, 43], 15: [11, 42], 8: [12, 41], 10: [13, 40], 37: [14, 39], 13: [15, 38], 44: [16, 37], 24: [17, 36], 6: [18, 35], 43: [19, 34], 29: [20, 33], 17: [21, 32], 7: [22, 31], 52: [23, 30], 46: [24, 29], 42: [25, 28], 40: [26, 27]]
Curva elíptica no singular para 53:  $y^2 = x^3 + 49x + 39$ 
Valor de a: 49
Valor de b: 39
Puntos en la curva: [(1, 6), (1, 47), (3, 1), (3, 52), (5, 12), (5, 41), (7, 6), (7, 47), (8, 25), (8, 28), (9, 19), (9, 34), (11, 1), (11, 52), (13, 8), (13, 45), (14, 17), (14, 36), (15, 11), (15, 42), (16, 19), (16, 34), (19, 12), (19, 41), (20, 3), (20, 50), (21, 10), (21, 43), (22, 23), (22, 30), (24, 26), (24, 27), (27, 2), (27, 51), (28, 19), (28, 34), (29, 12), (29, 41), (33, 4), (33, 49), (34, 26), (34, 27), (36, 12), (36, 32), (38, 13), (38, 40), (39, 1), (39, 52), (42, 17), (42, 36), (45, 6), (45, 47), (46, 25), (46, 28), (47, 18), (47, 35), (48, 26), (48, 27), (49, 16), (49, 37), (50, 17), (50, 36), (52, 25), (52, 28)]
El punto (1, 6, 1) está en la curva: True
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01>
```

P: 97

```
ments/ESCOM/BvoSemestre/CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio3.py"
Residuos cuadráticos y raíces cuadradas para 97: [0: [0], 1: [1, 96], 4: [2, 95], 9: [3, 94], 16: [4, 93], 25: [5, 92], 36: [6, 91], 49: [7, 90], 64: [8, 89], 81: [9, 88], 3: [10, 87], 24: [11, 86], 47: [12, 85], 72: [13, 84], 2: [14, 83], 31: [15, 82], 62: [16, 81], 95: [17, 80], 33: [18, 79], 70: [19, 78], 12: [20, 77], 53: [21, 76], 96: [22, 75], 44: [23, 74], 91: [24, 73], 43: [25, 72], 94: [26, 71], 50: [27, 70], 8: [28, 69], 65: [29, 68], 27: [30, 67], 88: [31, 66], 54: [32, 65], 22: [33, 64], 89: [34, 63], 61: [35, 62], 35: [36, 61], 11: [37, 60], 86: [38, 59], 66: [39, 58], 48: [40, 57], 32: [41, 56], 18: [42, 55], 6: [43, 54], 93: [44, 53], 85: [45, 52], 79: [46, 51], 75: [47, 50], 73: [48, 49]]
Curva elíptica no singular para 97:  $y^2 = x^3 + 34x + 38$ 
Valor de a: 34
Valor de b: 38
Puntos en la curva: [(1, 48), (1, 49), (3, 19), (3, 78), (4, 23), (4, 74), (6, 19), (6, 78), (9, 43), (9, 54), (11, 26), (11, 71), (15, 25), (15, 72), (16, 33), (16, 64), (17, 0), (20, 45), (20, 52), (21, 33), (21, 64), (22, 45), (22, 52), (23, 38), (23, 59), (24, 15), (24, 82), (27, 47), (27, 50), (28, 70), (29, 22), (29, 75), (30, 5), (30, 92), (33, 25), (33, 72), (34, 7), (34, 90), (35, 29), (35, 68), (36, 0), (37, 32), (37, 65), (42, 31), (42, 66), (43, 20), (43, 77), (44, 0), (46, 17), (46, 80), (48, 18), (48, 79), (49, 25), (49, 72), (52, 42), (52, 55), (54, 8), (54, 89), (55, 45), (55, 52), (56, 40), (56, 57), (58, 42), (58, 55), (60, 33), (60, 64), (62, 37), (62, 60), (63, 30), (63, 67), (64, 18), (64, 79), (65, 36), (65, 61), (70, 1), (70, 96), (71, 28), (71, 69), (72, 21), (72, 76), (75, 31), (75, 65), (76, 32), (76, 65), (77, 31), (77, 66), (78, 14), (78, 83), (79, 44), (79, 53), (81, 32), (81, 65), (82, 18), (82, 79), (84, 42), (84, 55), (85, 6), (85, 91), (86, 46), (86, 51), (86, 19), (88, 70), (91, 43), (92, 54), (93, 41), (93, 56), (94, 43), (94, 54), (96, 10), (96, 87)]
El punto (1, 48, 1) está en la curva: True
PS C:\Users\saiids\Documents\ESCOM\BvoSemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01>
```



P: 19

```
PS C:\Users\sahds\Documents\ESCOM\IVosemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01> & C:/Users/sahds/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/sahds/Docu
ments/ESCOM/IVosemestre/CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicios.py"
Residuos cuadráticos y raíces cuadradas para 19: 0: [0], 1: [1, 18], 4: [2, 17], 9: [3, 16], 16: [4, 15], 6: [5, 14], 17: [6, 13], 11: [7, 12], 7: [8, 11], 5: [9, 10]]
Curva elíptica no singular para 19:  $y^2 = x^3 + 15x + 15$ 
Valor de a: 15
Valor de b: 15
Puntos en la curva: [(3, 7), (3, 12), (4, 5), (4, 14), (5, 5), (5, 14), (6, 6), (6, 13), (7, 8), (7, 11), (8, 1), (8, 18), (9, 9), (9, 10), (10, 5), (10, 14), (12, 2), (12, 17), (14, 9), (14, 10), (15,
9), (15, 10), (16, 0)]
El punto (3, 7, 1) está en la curva: True
PS C:\Users\sahds\Documents\ESCOM\IVosemestre\CRYPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY\Practicas\Practica01>
```